

Predicting the Green Purchase Intention of Generation Z Café Consumers in Manila using Machine Learning Techniques

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Abstract—This study analyzes the green purchase intention (GPI) of Generation Z and their factors influencing decisions to patronize green cafés in Manila. Although concerns over the environment are rising, there is much less known about the characteristics of green cafés that determine generation Z consumers to select a green café. To fill this, utilize six machine learning techniques (Decision Trees, Random Forest, Gradient Boosting, K-Nearest Neighbors, XGBoost, and Support Vector Machines) on a dataset of 444 Gen Z consumers. The results showed that subjective norms, green perceived quality, and social influence were significant predictors, while environmental awareness and eco-social benefits were weak factors in this process. In the combined model, random forest and support vector machine achieved the highest overall model accuracy of 81% when predicting consumer purchasing behavior.

Keywords—green café, sustainability, coffee shops, generation Z, green purchase intention, machine learning, modelling, artificial intelligence, analytics

I. INTRODUCTION

environment has made a small yet significant change in consumer behavior, which has resulted in rising demand for products and services that reduce ecological footprints. Studies in the Philippines report that 75% of consumers try to look for brands that offer eco-friendly products [1]. The increase of environmental awareness of Generation Z in Manila allowed the local shops to adopt eco-friendly practices such as waste reduction, biodegradable packaging, energy efficient operations, and responsible green sourcing that resonates with the increasingly environmentally aware consumer base [2].

Research shows that 80% of Filipino Gen Z consumers try to reduce their environmental impact that might be to an extent formed by their values which leads to their decisions playing an important role in the future market trend [3]. This

trend of consumer purchases based on personal values is one example of green consumption, where consumer purchases are motivated by a sense of responsibility for the health of the environment and society. Given Gen Z's considerable market share [4], understanding and predicting how they, as café consumers, react to green practice is thus crucial for promoting sustainable consumption and business innovation to stay competitive and relevant.

Despite the considerable environmental awareness of Gen Z, a significant "say-do gap" persists between their stated pro-environmental intentions and their actual purchasing behavior. This lack of similarity challenges businesses, as the factors translating general eco-awareness into a specific preference for green coffee shops remain unknown [5, 6]. Furthermore, although Gen Z's sustainable concerns are well-documented, there is a lack of local predictive study that specifically identifies the drivers of Gen Z's purchase intention in a developing country like the Philippines.

By using machine learning tools to identify and predict Generation Z's thinking, social, and environment drivers, this study can address the gap the Philippines is facing. The study implemented six supervised machine learning algorithms such as Decision Trees (DT), K-Nearest Neighbors (KNN), Support Vector Machines (SVM), Random Forests (RF), Gradient Boosting (GRB), and XGBoost (XGB) for classification and prediction.

II. LITERATURE REVIEW

A. Sustainable Consumption in the Philippines' Food and Beverage Industry

Green consumption in the Philippine food production reflects the growing shift towards environmentally responsible business practice and consumer behavior. While many

restaurants and cafes began adopting eco-friendly practices such as using “eco-friendly packaging”, “energy-efficient facilities and waste reduction initiatives”, customer awareness and understanding may become a problem [7]. Although customer awareness of sustainability efforts remains limited, green practices have proven to create meaningful environmental and operational benefits in the food service industry. Restaurants in the Philippines that adopted AI-driven food waste reduction systems were able to minimize surplus and resource waste, showing how technology-supported sustainability contributes to environmental protection [8]. Moreover, adoptions of no single use of plastic policies in local restaurants have shown significantly less plastic waste generation and this promotes a green environment responsibilities on businesses [9]. The connection between sustainable practices and customer behavior serves as a basis for identifying Generation Z’s customer purchase intention towards eco-friendly cafes establishments in Manila shows that sustainability is not just a trend, but rather an effective improvement for the Philippine food and beverage industry.

B. Green Purchase Intention

Green purchase intention is crucial aspect of sustainability in business firms, as well as improving customer relationships. Several factors have been identified on green purchase intention, such as price value, green innovation, and green perceived quality, among others. While price remains a factor, with a positive outlook, Generation Z consumers show a greater responsiveness to sustainability cues such as eco-labels compared to older generations and are more likely to purchase eco-friendly products [10]. Moreso, green establishments that demonstrate clear eco-friendly practices such as reduced plastic use, reusable packaging, responsible green sourcing, and visible sustainability initiatives which signals credibility and aligns with Generation Z’s values strengthens their purchasing intentions [11].

Influence of Environment (IE) indicates that customer potential for adopting green cafes into their routine purchase behavior will depend on the access, visibility, and availability of these cafes for them [12]. Eco-Social Benefits (ESB) identify the positive, socially constructive impact customers receive when a green cafe minimizes waste, sources its supplies ethically, and becomes involved in their community [13]. Green Perceived Quality (PQ) describes how customers assess the reliability, quality, and environmental worth of green cafe goods and services [14]. Purchase Motivation (PM) It reflects the consumer's personal urges, feelings, and wishes that specifically lead them to look for and purchase environmentally friendly products [15]. Environmental Awareness (EA) includes knowledge and interest of individual(s) toward environmental issues, resulting in pro-environmental actions and attitudes [16]. Green Self-Identity (SI) is the extent to which consumers view and refer to themselves as environmentally responsible individuals who associate their self-concept with "green" issues and activities [17]. Influence of Pricing (IP) shows how perceptions of affordability and product value determine purchase intention

and buying behavior in a café environment [18]. Subjective Norms (SN) are the social expectations of friends or society on the green practices and the influence that has on the intention to buy green products [19]. Green Innovation (I) is the development and provision of sustainable products, services, and processes that optimize consumer satisfaction and encourage repeat sustainable patronage. Green Purchase Intention (PI) gauges the extent to which individuals can be influenced to make eco-friendly purchases based on their internal attitudes and behavioral control.

III. MATERIALS AND METHODS

A. Survey Development

The survey instrument was designed into two main sections, based on prior studies. The first part collected the profile of the respondent, while the second part examined the factors influencing their intention to purchase in an eco-friendly café using a 7-point Likert scale. Consistent with ethical research conduct, a screening question, data protection confidentiality agreement, and brief objectives of the research were included at the beginning of the survey.

B. Data Gathering and Participants

The researchers conducted a purposive sampling method to achieve the goal of the study. This non-probability sampling method is suitable as it enables the collection of data from participants whose characteristics align closely with the specific parameters and goals of the research. For this study, the criteria required that target respondents belong to Generation Z and have experience visiting and purchasing from eco-friendly coffee shops in Manila. Data collection was conducted using a survey developed in Google Forms, which was shared through different social media channels that focus on café-related communities. To supplement this digital distribution, researchers also extended in-person invitations to patrons at green cafés within Manila, Philippines.

Demographic profile of the 444 respondents who are primarily female (55.6%) and are generally within the age category of 19 to 28 years, which locates the respondents as young adult consumers regularly dining at cafés likely to have a high exposure to sustainability-focused consumption. The majority (79.1%) were college or university students, with smaller percentages identifying as young adults in the workforce. Most of the participants indicated visiting cafés at least once a month and indicated similar frequency for frequency (27.0% - twice or more per week, 24.3% - once per week, and 18.2% - once every two weeks). Similarly, survey results indicated that the expenditure level was in the range of ₱100 to ₱199 (42.3%) and ₱200 to ₱299 (35.1%). Also, most respondents (38.3%) had allowances below ₱5,000 and the most consisted of the (27.9%) with allowances below ₱5,000 to ₱9,999, which suggested a more cost-conscious group. The profiling shows how Gen Z café patrons are relevant for understanding the importance of sustainability in eco-friendly coffee shop consumers without expectations of increases for sustainability.

C. Data Preprocessing

In data preprocessing, Python was used to evaluate the completeness and consistency of the data. This included checking for missing data and dropping them, eliminating inputs that fell outside of the study's limitations, and verifying all responses were deemed suitable for modeling. Upon cleaning the data, the categorical variables used for analysis, such as demographics, were changed to, or reencoded to, one-hot encoding, while Likert-scale items were reencoded to ordinal data. This procedure of the data processing phase was relevant to adequately prepare for analysis and predictive modeling.

D. Model Development

The study tested several machine learning algorithms, including DT, RF, GRB, XGB, KNN and SVM. To train and test the model efficiently, the data was split into two sets, 80% for training, and 20% for the test model. All the input variables were used to calculate mean scores and create one continuous measure of items which mapped to each factor (IE, ESB, PQ, PM, EA, SI, IP, SN, and I). The output variable (PI) was conveniently converted to a binary variable meaning that any score indicating a mean score of above 5.79 was then classified as high purchase intention. The whole modeling procedure is shown in Figure 1.



Fig. 1. Modeling Process

To further improve the models' predictive accuracy, GridSearchCV was adopted to optimize hyperparameters. This approach is combined with the K-fold cross-validation procedure to assess thoroughly each model's performance.

TABLE I. HYPERPARAMETER SETTINGS

Algorithm	Hyperparameter	Value
Gradient Boosting	learning_rate	0.05
	n_estimators	50
Random Forest	max_depth	10
	min_samples_leaf	4
	min_samples_split	2
	n_estimators	50
SVM	C	1
	gamma	0.01
	kernel	rbf
K-Nearest Neighbors XGBoost	n_neighbors	7
	booster	gbtree
	learning_rate	0.01
	max_depth	3
	n_estimators	100
Decision Tree	criterion	gini
	max_depth	None
	min_samples_leaf	2
	min_samples_split	10

E. Performance Evaluation

To predict the green purchase intention of Generation Z towards eco-friendly cafes, machine learning tools were

utilized. To enhance prediction accuracy and interpret the key factors that influence eco-friendly cafe choices among Generation Z consumers, Predictive models were implemented, particularly bagging and boosting algorithms. Accuracy, precision, recall, F1-score, and the area under the receiver operating characteristic curve (AUC-ROC) was used to evaluate the predictive performance of the machine learning models.

IV. RESULTS AND DISCUSSION

A. Machine Learning Models Performance

The application of all six machine learning models is shown in Table II. Random Forest and GB showed strong performances, GB had the highest Recall (0.94) among all models, while Random Forest represented balanced Precision (0.79) and Recall (0.90), achieving an F1-score of 0.84 and had accuracy of 0.81 for both models. SVM with an Accuracy of 0.78 and a low Hamming Loss (0.22). The KNN algorithm had a moderate performance, having an accuracy of 0.76, strong Recall (0.90), and an F1 score of 0.81, though with a higher Hamming Loss of 0.24. A very similar outcome in Accuracy (0.76) was shown by the Decision Tree model, which had the lowest Recall (0.80), the lowest F1 score (0.79), and the highest Hamming Loss (0.24), making it the least reliable.

TABLE III. PERFORMANCE OF MACHINE LEARNING MODELS

Model	Accuracy	Precision	Recall	F1-score	Hamming score	Kappa score	AUC-ROC
RF	0.81	0.79	0.90	0.84	0.19	0.72	0.90
GB	0.81	0.77	0.94	0.84	0.19	0.75	0.88
XGB	0.80	0.79	0.86	0.82	0.20	0.73	0.87
SVM	0.78	0.75	0.90	0.81	0.22	0.75	0.87
KNN	0.76	0.73	0.90	0.81	0.24	0.76	0.83
DT	0.76	0.78	0.80	0.79	0.24	0.74	0.78

Figure 2. shows that all models RF, SVM, GB and XGBoost have good performance with AUC scores ranging from 0.87 to 0.90. These specify that the leading models are able to identify green purchases intention outcomes roughly 87% - 90% of the time, with RF having the highest AUC score at 0.90. KNN also demonstrates predictive performance with an AUC score of 0.83, while DT is lowest, with an AUC score of 0.78, but still performs better than random guessing. Overall, the ROC curves show that all models' prediction performance is definitively above chance level, the ROC curves for the top models (RF, SVM, GB and XGBoost) indicate an even more robust predictive performance, given their high true positive rates against their false positive rates, while avoiding misclassification, even for DT which performed lower than other models but still demonstrated predictive performance.

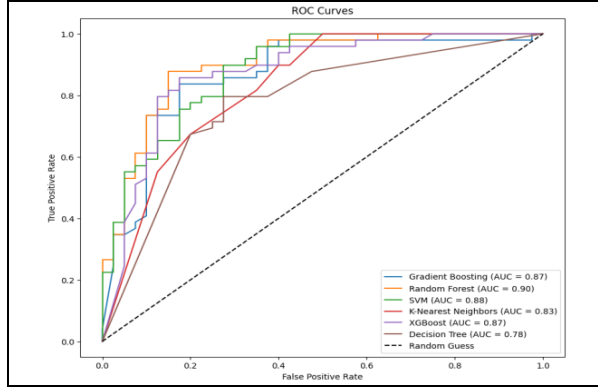


Fig. 2. ROC Curves of Machine Learning Models

B. Machine Learning Models Feature Importance Plots

The importance of the features used for all six models is shown in Figure 3. To identify and rank the most impactful predictors influencing model output, two measures were used: scikit-learn's feature importance and permutation methods. These plots show the extent to which a certain predictor impacted a model's ability to predict Gen Z consumers' green purchase intention. Gen Z's purchase intention is illustrated in Figure 5. across KNN, XGB, GB, RF, SVM, and DT models employing different rankings that vary in feature importance comprising three distinct tiers. SI is ranked the highest (average 1.83) due to its persistent high placement in all models. SN is ranked next (2.33), while PQ (2.50) was determined to be of importance although it ranked low in KNN. PQ took first place in both DT and the ensemble models XGB and GB. A noticeable gap can be observed between the top tiers and PM, which holds a stable mid-range rank (4.67) and a moderate impact. Conversely, the rankings of I, IE, and IP (5.83 - 6.33) are considerable, while ESB (7.50) and EA (7.67) rank lowest and have little impact on Gen Z purchase intention. Overall, ensemble models, which include GB, RF, and XGB, highlight PQ, SI, and SN. Meanwhile, margin-based SVM, and distance-based KNN methods are much more focused on SI and SN.

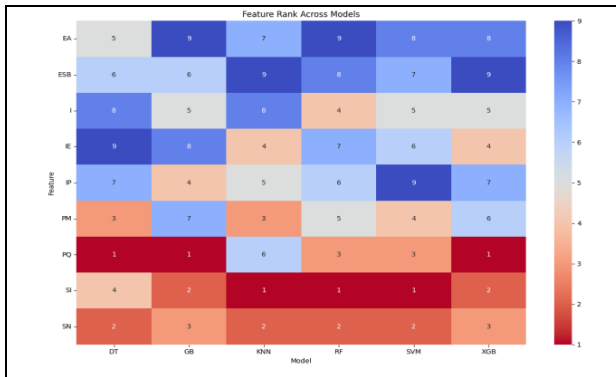


Fig. 3. Feature Importance Heatmap by Model

C. Discussion

As sustainability becomes a growing emphasis, consumer decisions have begun to shift. For Gen Z, this means their decision to purchase from a cafe is mainly contingent on

different factors. The study examined the predictors of green cafe purchasing intention and measured the appropriateness of machine learning models. All six models had acceptable performance measuring over 76% accuracy; Random Forest and SVM were the two strongest classifiers (AUC=0.90, 0.88) historical evidence - which makes sense, as previous research indicated the reliability of RF and SVM for complex classification algorithm educational tasks. Although Decision Tree ranked lowest with an AUC-ROC score (0.78), Decision Tree achieved acceptable predictive capability too, which ultimately demonstrates the trajectory of applying machine learning in modeling consumer behavior in sustainability contexts. Green Self-Identity (SI) reliably ranked as the number one or first ranked predictor across models, immediately followed by Subjective Norms (SN) and Green Perceived Quality (PQ); Gen Z prefer green cafes when they identify with sustainability, or it is a fit for their self-concept or identity, especially when they perceive the fit is high quality.

The mid-ranked predictors were Purchase Motivation, Green Innovation, Influence of Environment, and Influence of Pricing, which had a moderate influence. Green Self-Identifications were always ranked high as the primary motivator of green purchase intention in Random Forest, Gradient Boosting, XGBoost, KNN, and SVM Models, Subjective Norms would have significant influence over Gen Z's likelihood of purchasing from a green cafe. The high influence of Subjective Norms in the results supports Gen Z's susceptibility to peer influence while the Perceived Quality variable indicates that green cafes can provide sustainable practices with adequate product quality if they are to significantly impact consumer choices. The Environmental Awareness and Eco-Social Benefits variables would both be the weakest predictors of green purchase intention and have been ranked as the weakest predictor of intention across all five models. According to the SHAP results, Subjective Norms were the most influential predictor between .090-.124, followed by Green Perceived Quality at .082-.123 and Green Self-Identity at .078-.091. Eco-sustainable behavior and Environmental Awareness were the least influential predictors with values closer to zero. The SHAP analysis also enhanced interpretability of the model by indicating that Subjective Norm, Green Perceived Quality, and Green Self-Identity had the highest marginal effects on prediction, while mid- and low-influence predictors had lower SHAP values which support the stability of the model and the consistency of predictor influence compared to other algorithms. Since these two variables examine general eco-awareness and general benefits, it is insufficient to understand Gen Z's purchase motivation from eco cafes. The data analysis indicates, instead, that green purchasing is largely driven by personal identity, social norm, and quality perceptions. This is consistent with results from the recent research that showed environmental awareness does not contribute to sustainable behavior alone, unless the eco-awareness is cultivated by culturally normative expectations or through social cues.

V. CONCLUSION

This paper investigated the predictors of Generation Z's green café purchase intention using machine learning methods. It was found that Green Self-Identity had the strongest influence on purchase intention. Subjective Norms and Green Perceived Quality were also strong predictors, which is consistent with the argument that the social norms and the expected sustainable product quality were important predictors. Variables of Purchase Motivation, Green Innovation, Influence of Environment and Influence of Pricing were weak-to-moderate predictors of the purchase intention. Environmental Awareness and Eco-Social Benefits were the lowest predictors, which suggested that the general environmental problem knowledge and eco-benefits awareness are less influential on purchase intention unless supplemented by any profound reason with personal or social aspects. The SHAP analysis also enhanced interpretability of the model by indicating that Subjective Norm, Green Perceived Quality, and Green Self-Identity had the highest marginal effects on prediction, while mid- and low-influence predictors had lower SHAP values which support the stability of the model and the consistency of predictor influence compared to other algorithms.

Some limitations of the study indicated that future research is still needed. First, the present study was based on the survey data on the perceptions of Generation Z. It may limit the predictive models' ability to predict the green café purchase intention consistently over the times. Future work can improve this by developing a more complete model based on the economic indicators and historical transaction data. Second, this study focused on the generation-z participants, leading to the exclusion of further participants such as millennials and older consumers. Future investigations may be able to conduct comparison studies between generation z and millennials, or other consumer groups. In addition, advanced feature engineering techniques were not employed in a way that may have led to an optimized model performance.

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